

Myasthenia Gravis

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Outlines

- Definition of Myasthenia Gravis
- Causes of Myasthenia Gravis
- Pathophysiology
- Clinical manifestations of Myasthenia Gravis
- Diagnostic tests of Myasthenia Gravis
- Medical treatment of Myasthenia Gravis
- Nursing management of Myasthenia Gravis



Pre test

Mention causes
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Definition;

Produces sporadic but progressive weakness and abnormal fatigue in striated (skeletal) muscles.

This weakness and fatigue are exacerbated by exercise and repeated movement but improved by anticholinesterase drugs.

Usually, myasthenia gravis affects muscles innervated by the cranial nerves (face, lips, tongue, neck, and throat), but it can affect any muscle group.

Causes of the disease

Myasthenia gravis has an unpredictable course that includes

periods of exacerbation and remission.

Drug treatment has improved the prognosis and allows patients to lead relatively normal lives, except during exacerbations.

However, if the disease involves the respiratory system, it can be life-threatening

The cause of myasthenia gravis isn't known; however, it commonly accompanies autoimmune and thyroid disorders.

Pathophysiology

The patient's blood cells and thymus gland produce antibodies that block, destroy, or weaken the neuroreceptors that transmit nerve impulses, causing a failure in transmission of nerve impulses at the neuromuscular junction.

In myasthenia gravis, antibodies attach to the ACh receptor sites.

They block, destroy, and weaken these sites, leaving them insensitive to ACh, thereby blocking neuromuscular transmission.

Clinical manifestations

Common signs of myasthenia gravis include:

according to affected nerve

1. Gradual, progressive skeletal muscle weakness and fatigue that worsens during the day
2. Weak eye closure, ptosis, and diplopia
3. Blank, masklike facial expression
4. Difficulty chewing and swallowing
6. lateral motion of the head
7. Symptoms of respiratory failure if respiratory muscles are involved.

Diagnostic tests



The Tensilon test confirms the diagnosis by temporarily improving muscle function after an I.V. injection of neostigmine.



Long-standing ocular muscle dysfunction, however, may not respond.



Electromyography helps differentiate nerve disorders from muscle disorders.



Nerve conduction studies test for receptor antibodies.

Medical treatment

Treatment is symptomatic.

- 1. Anticholinesterase drugs
- 2. Corticosteroids may help to relieve symptoms.
- 3. Plasmapheresis.
- 4. One with thymomas requires thymectomy, which may lead to remission in adult-onset myasthenia gravis.



In crises Acute exacerbations that cause severe respiratory distress necessitate emergency treatment.



Tracheotomy, or positive-pressure ventilation, or vigorous suctioning to remove secretions usually yield improvement in a matter of days.



Anticholinesterase drugs aren't effective during myasthenic crisis, so they're discontinued until respiratory function begins to improve.

Nursing management;

- Establish an accurate neurologic and respiratory baseline
- Help remove secretions as they accumulate
- Be alert for signs of an impending crisis (increased muscle weakness, respiratory distress, and difficulty talking or chewing)
- For the best results, administer drugs at evenly spaced intervals and on time, as ordered.
- Be prepared to give atropine for anticholinesterase overdose or toxicity.

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- Plan periods of exercise, meals, patient care, and daily activities to take advantage of peaks in the patient's energy level.
 - Provide soft, solid foods instead of liquids to reduce the risk of choking.
 - Always sit the patient up to eat.
 - Encourage the patient to take an active role in deciding about his care.
 - Evaluate the patient.
 - Look for normal vital signs, evidence of adequate hydration and normal elimination, skin that's free from sores or problems, and an optimal capacity for activity.

Myasthenia gravis teaching tips



1. Help the patient plan daily activities to coincide with energy peaks.



Stress the need for frequent rest periods throughout the day.



Emphasize that periodic remissions, exacerbations, and day-to-day fluctuations are common.



2. Teach the patient how to recognize adverse effects and signs of toxicity of anticholinesterase drugs (headaches, weakness, sweating, abdominal cramps, nausea, vomiting, diarrhea, excessive salivation, bronchospasm) and corticosteroids.



Warn him to avoid strenuous exercise, stress, infection, and unnecessary exposure to the sun or cold weather.



Caution him to avoid taking other medications without consulting his primary health care provider

Post test

Mention causes
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Mention3
symptoms of
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ANY QUESTION



THANK YOU
